dune-rover executive brief

Executive brief: predicting slip and traversability of wheeled robots on Arabian dunes

Huda Mueen, `hudasol/dune-rover`, 2026-10-28 (benchmark of stages 25 and 26 added). Generated by `scripts/build_report.py`; every number comes from a result file or an evidence row in `research/evidence.csv`. **This is a proposal and a research summary. No agreement, partnership or endorsement exists with any institution, and none is implied.** Heat and dust limits are the subject of the separate repository `hudasol/dhill`. No cost is given.

Labels: **PUBLISHED** (read in a retrieved source), **SIMULATED BY ME** (output of this repository's code; simulation only, never field evidence), **PLANNED** (not run).

1. Question and Gulf relevance

How much does a small wheeled robot slip on a dune slope, can it be predicted, and how fine must a terrain map be to plan on Gulf dunes? No published slip against slope dataset exists for a wheeled robot on Arabian dune sand; the only Arabian field item is a qualitative report of a rover stuck in loose sand in Oman (FLD-042). Open elevation data for the region are about 30 m.

2. What is published (PUBLISHED; 510 verified rows from 176 sources)

- **Earth dune slope curve.** At Dumont Dunes, California, a 340 kg test rover went from below 50 % slip at 12.5 degrees to about 90 % at 20 degrees, without soil parameters (FLD-HV13-1, FLD-HV13-2).
- **Field slip is large and patchy:** Spirit 15 to 20 % at 19 degrees and 95 % at 16 degrees on look alike terrain (FLD-011, FLD-012); Perseverance above 94 % at about 20 degrees (FLD-025).
- **Arabian sand:** grain size, sorting, mineralogy and density of UAE, Qatari, Saudi and Omani dune sands are published (SND rows); no friction angle or slip face angle measured on UAE dunes was found.

3. What was tested (SIMULATED BY ME; every main test pre registered in `docs/PREDICTIONS.md`, post hoc analyses labelled)

- **Slip models against published curves.** On 49 published curves from 10 sources, three model families (exponential, Mohr Coulomb, logistic) calibrated on other studies predicted a held out study with median slip errors of 0.20 to 0.22; none met the bar of 0.15 fixed beforehand, and an empirical curve did as well as the physics. A fourth, Bekker Wong, failed on the 8 first round curves it applies to (0.20); on the expanded set it reached 0.41 with coverage 0.46 on 33 curves. Two families built on the granular scaling laws did worse (0.31 to 0.34).
- **Independent simulator.** Project Chrono with published soils gives 0.21 slip at 12.5 degrees for a Scarecrow sized wheel, against about 0.49 on the Dumont curve; calibrated on that curve up to 12.5 degrees, it gave no usable prediction of 15 to 20 degrees (the run was not numerically stable there in post hoc checks).

- **Terrain map against real lidar.** On 6 North American dune fields (USGS one metre lidar), 16.3 % to 28.6 % of the cells a 30 m three class map called TRAVERSABLE contained steep slip faces (1.2 % to 3.6 % at 10 m).
- **Arabian dunes from space.** Over Liwa (A), the Rub' al Khali megadunes (B) and the linear dunes (C), 71 %, 47 % and 67 % of the ICESat-2 segments steeper than 28 degrees fell in cells the 30 m map calls TRAVERSABLE, against 93 %, 92 % and 98 % of all segments; keeping only steep crossings confirmed by a neighbouring segment and no steeper than 35 degrees (post hoc), 52 %, 29 % and 49 %. The ICESat-2 method failed its own validation against lidar, so these are indications, not a validation.
- **Closed loop.** In simulation, an online slip estimate from odometry and position fixes (median error about 0.029, set by the assumed position noise) and replanning on it showed no effect on entrapment distinguishable from a stop and retry control over 5 slip fields (post hoc); the true rover is itself a model.
- **Baselines on six lidar dune fields (stage 25).** At 10 m every face detector except roughness ranks face cells well (mean AUC 0.93 to 0.97). At 30 m a cubic interpolation of the coarse surface ranks them best or tied best (mean AUC 0.82), gradient boosting and a convolutional network trained on lidar of the other fields follow (0.81, 0.79), and this repository's synthetic trained detector (0.74) beats only plain slope (0.70). On the real Copernicus GLO-30 product every method falls to 0.65 to 0.73.
- **The planner against baselines (stage 26).** With the shipped planner and 600 origin and destination pairs scored on face patches in the lidar, mission success on 30 m maps is 6 % with the slip model alone and at most 11 % with any hazard layer a rover could compute; on 10 m maps it rises to 34 % with the best layer; a perfect cell label (at least 10 steep posts per cell) gives 46 % at 10 m and 59 % at 30 m, and a hard gate on a face map that marks every cell holding a face post finds a route (face free by construction) for 87 % of pairs at 10 m, 83 % at 30 m (post hoc) and 100 % at 5 m. Conference paper: `paper/paper.pdf`.
- **Planning (earlier stages).** On the first three fields no policy was safer than naive planning on 30 m maps against lidar truth, and with 10 m maps a conservative policy failed less often than naive at two of three sites on the pairs it could route (post hoc comparison); on three further fields the conservative policy was somewhat safer at 30 m, but most routes still failed at two of them.
- **Arabian maps.** With slip parameters calibrated on published curves, UNKNOWN at 30 m is 62.8 % (Liwa), 44.3 % and 46.0 % (Rub' al Khali), against 8.6 %, 10.9 % and 2.5 % under the original assumed ranges.

4. Negative results

- No slip model transfers between vehicles and soils with useful accuracy; two implementations of the same Bekker and Janosi laws disagree by more than field scatter.
- A 30 m map labels 36 % to 80 % of the cells that contain a steep face as TRAVERSABLE (7 % to 33 % at 10 m); the slip face detector works much better at 10 m than at 30 m.
- At 30 m no hazard model, ours included, plans many face free routes; the better layers roughly halve the length of route on faces (50 to 19 m per km, post hoc). Given a face map made from fine data, a hard gate on the 30 m grid still finds face free routes for 83 % of pairs (post hoc), so safe routes need face information from a drone survey or onboard sensing more than a finer planning grid.
- The planned pass or fail rule accepts a model that collapses above 20 degrees in 53 % of synthetic replicates; stricter options reject a correct model more than half the time.

5. Error rate of the evidence table

Every row was re checked against its full text by a separate agent. Of the 323 rows added on 2026-10-20, 30 had a material error before correction (9.3 %, Wilson 95 % interval 6.6 to 12.9 %); every material error is corrected. The 16 rows added on 2026-10-21 had 0 material errors. A human spot check and a second rater pack are prepared.

6. The first field experiment (PLANNED)

A slope ladder with one small four wheel rover (5 to 25 kg) on one dune stoss slope (bins 0 to 20 degrees, denser between 10 and 20), a 12 run pilot, RTK or drone slope, soil tests at every station, calibration up to 10 degrees and predictions committed before steeper runs, plus a drone survey of the dune at 10 m or finer to test the map on Arabian terrain (`docs/FIELD_VALIDATION_PLAN.md`).

7. The ask

A supervised dune site, a small wheeled rover, RTK GNSS or a drone survey, soil tests and recovery support; work packages and fallbacks are in `docs/PROPOSAL_ROBOTICS.md` and `docs/PROPOSAL_EDGE.md`. Success means a complete dataset and an honest comparison with the committed predictions, even if the model fails. The deliverable is the Dune Rover Planner (`docs/PLANNER.md`), already working in simulation with a resolution gate, calibrated to the partner's rover and checked on a surveyed Gulf dune.